

Yogith Ramanan

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EDUCATION

DePaul University

Master of Science in Computer Science

Chicago, IL, USA

January 2025 – Present

Sri Krishna College of Engineering and Technology

Bachelor of Engineering in Computer Science

Coimbatore, Tamil Nadu, India

July 2020 – May 2024

EXPERIENCE

GURU Information Technology Services

Product Development and Support Engineer Intern

January 2024 – June 2024

Chennai, Tamil Nadu, India

- Supported product stability for customer-facing applications by resolving UI defects and production incidents across HTML, CSS, and JavaScript, directly reducing user-reported issues and improving end-user experience for enterprise clients.
- Accelerated incident resolution by diagnosing root causes through log analysis, delivering targeted patches, and documenting remediation steps, minimizing repeat occurrences and reducing mean time to recovery (MTTR) across production systems.
- Authored detailed defect reports and validation summaries using structured debugging workflows; collaborated cross-functionally with engineering teams to enforce quality gates and improve deployment readiness end-to-end.

Gateway Software Solution

Full Stack Intern

July 2022 – September 2022

Coimbatore, Tamil Nadu, India

- Contributed to full-stack product delivery for a software consultancy serving multiple clients, building interactive dashboards and data-driven interfaces that translated backend data into actionable business insights for end users.
- Improved front-end reliability and development velocity by maintaining front-end, back-end, and database layers in parallel, reducing regressions across releases and ensuring consistent feature delivery against evolving client requirements.
- Applied GitHub-based version control workflows for branching, code review, and team collaboration; enforced deployment readiness standards that improved review efficiency and reduced integration conflicts across the development lifecycle.

RESEARCH EXPERIENCE

Carbon-Aware Scheduling for Multi-Region AI Inference | *Python, Cloud Systems, Simulation* Jan 2026 – Mar 2026

- Designed a Python discrete-event simulation evaluating 8 cloud routing policies across 5 AWS regions and 3 AI workloads (BERT-base, BERT-large, ResNet-50), processing 33,600 requests over 7 days using real-world carbon intensity and latency data.
- Delivered production-viable impact at enterprise scale: the Constrained Hybrid policy cuts per-request carbon cost by 54.8% across millions of daily inference calls, translating to hundreds of megatons of avoidable CO₂ at industry scale, while adding only 24.8 ms average latency overhead and maintaining 0% SLO violations, making it directly deployable in real cloud load balancers with ~50–100 lines of routing logic.
- Built a novel Adaptive Hybrid controller using closed-loop P95 latency feedback to dynamically adjust routing weights per workload, eliminating manual parameter tuning and achieving 29.7% carbon reduction at 4.53% SLO violation rate.
- Produced an enterprise decision framework mapping 5 deployment scenarios (strict SLO, moderate SLO, relaxed SLO, ultra-strict, batch async) to recommended policies, with Pareto trade-off analysis across all 8 configurations for cloud architects.

PROJECTS

AgentMind — Autonomous Multi-Agent AutoML System | *LangGraph, FastAPI, React, Groq, ChromaDB* Apr 2026 – Jun 2026

- Targets the \$1.1B+ AutoML market as a free open-source alternative to DataRobot (\$2,500/month), enabling business analysts and enterprises to obtain a trained, optimized ML model in under 60 seconds, directly reducing the barrier to AI adoption for non-technical teams.
- Demonstrated measurable ML impact across 4 real-world datasets: 88.5% accuracy on heart disease prediction and 85.3% on a 145,460-row weather dataset, compressing what typically takes a data science team days of model selection and tuning into under 60 seconds, with full SHAP explainability for business-interpretable decisions.
- Built a parallel ML pipeline training 5 models simultaneously (Random Forest, XGBoost, LightGBM, CatBoost, Logistic Regression) using ThreadPoolExecutor with Optuna Bayesian tuning and SHAP explainability; shipped full-stack React + Vite frontend with FastAPI backend, deployed live on Vercel and Render with CI/CD auto-deploy on every GitHub push.

Multimodal AI Platform for Drug Candidate Prioritization | *PyTorch, FastAPI, Docker, MLflow* Jan 2026 – Mar 2026

- Addresses a critical bottleneck in pharmaceutical R&D where drug discovery costs exceed \$2.5B per approved compound; the platform accelerates candidate screening using ML-driven prioritization, offering a reproducible, cost-effective alternative to manual assay workflows.
- Quantified model impact through a rigorous three-way evaluation (ROC-AUC, PR-AUC, F1, balanced accuracy + rank correlation): the graph-tabular fusion model outperformed both the logistic regression baseline and Random Forest fusion, demonstrating that incorporating molecular graph structure meaningfully improves candidate prioritization quality, representing the difference between surfacing the right drug compound early vs. costly late-stage failures.
- Implemented deterministic candidate ranking with rule-based descriptor penalties, confidence scoring, and reason codes for auditable decision support; containerized FastAPI service with `/predict`, `/rank`, `/batch-rank` endpoints, pytest + GitHub Actions CI, Docker Compose packaging, and Prometheus-compatible `/metrics` instrumentation.

Serverless Document Processing System | *AWS Lambda, S3, Textract, DynamoDB, API Gateway* Sep 2025 – Oct 2025

- Solves enterprise document digitization at scale by replacing manual OCR workflows with a fully serverless, event-driven pipeline on AWS managed services, enabling zero-ops document processing that scales from 1 to millions of files without infrastructure overhead.
- Delivered a zero-infrastructure-management system capable of processing documents end-to-end, from S3 upload to structured text extraction and queryable API, with automatic scaling to any volume, eliminating dedicated OCR servers and reducing operational cost to near-zero with pay-per-use Lambda pricing.
- Developed RESTful API Gateway endpoints with API-key authentication, least-privilege IAM controls, and structured 4xx/5xx lifecycle handling; designed DynamoDB schema for document metadata lifecycle (UPLOADED → PROCESSED) with CloudWatch metrics for latency, failure observability, and full end-to-end traceability.

MealSnap AI — Image-Based Food Calorie & Nutrition Tracker | *TensorFlow, MobileNetV2, Streamlit* Jun 2025 – Aug 2025

- Targets the \$220B+ global digital health and nutrition tracking market; deployed as a live public web app serving real users, demonstrating production ML delivery with immediate business applicability in consumer wellness and chronic disease management.
- Achieved production-grade model performance: 85.3% top-1 and 96.2% top-3 accuracy on 101 food classes at under 250 ms CPU inference latency, with a multi-signal OOD detection system (confidence threshold, entropy, top-3 spread) achieving 95%+ non-food rejection, making the app reliable enough for real users to trust daily nutrition decisions on.
- Built human-in-the-loop correction workflow with portion-aware nutrition scaling (USDA FoodData Central, 124 foods) and meal history CSV export; optimized production performance via `@st.cache_resource` model caching, abstention logic for low-confidence predictions, and responsive UI deployed on Streamlit Community Cloud.

A2A Java SDK — Agent-to-Agent Communication Platform | *Java, Spring Boot, REST APIs* Jan 2025 – Mar 2025

- Addresses a growing need in enterprise agentic AI architectures where standardized inter-agent communication is critical; the SDK reduces integration time for teams building multi-agent systems by providing typed, reusable client abstractions out of the box.
- Reduced agent integration complexity for multi-agent systems by providing a strongly-typed, reusable SDK replacing ad-hoc HTTP wiring with standardized abstractions that cut boilerplate, enforce consistent error contracts, and give developers observable, traceable inter-agent calls out of the box with zero additional instrumentation.
- Improved integration reliability through rigorous payload validation, edge-case handling, and standardized error handling following SOLID principles; packaged with Maven, enforced unit testing with JUnit, and applied consistent API design patterns for drop-in usability across multi-agent orchestration frameworks.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, Bash

Frontend / UI: React, Vite, HTML5, CSS3, Streamlit

Backend / APIs: FastAPI, Flask, Django, Spring Boot, Node.js, REST APIs, OpenAPI, GraphQL

Cloud (AWS): EC2, S3, Lambda, RDS, API Gateway, CloudWatch, SageMaker, EKS, IAM, VPC, Textract, EventBridge, AWS Batch

DevOps / MLOps: Docker, Docker Compose, Kubernetes, Terraform, GitHub Actions, CI/CD, Jenkins, MLflow, Linux

AI / ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Optuna, Bayesian Tuning, Feature Engineering, Transfer Learning, GCN, Multimodal AI, Parallel Model Training

Agentic AI: LangGraph, LangChain, Multi-Agent Orchestration, RAG, ChromaDB, Groq API, OpenAI API, Prompt Engineering, Chain-of-Thought Reasoning

Data / Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, ChromaDB, Pandas, NumPy, Spark

Cheminformatics: RDKit, SMILES, Morgan Fingerprints, Molecular Descriptors, Drug Candidate Prioritization

Observability / Testing: Prometheus, CloudWatch, pytest, JUnit, Structured Logging, Integration Testing

Principles: OOP, SOLID, Design Patterns, Agile/Scrum, System Design, Microservices, Distributed Systems